

BUILDORA

AI-Powered Construction Workflow Automation

Product Requirements Document · Version 1.0

UM HACKATHON 2026 - Domain 1: AI Agentic Workflow Automation

TEAM NAME

Dang Wangi

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Project Details

Project	Buildora
Hackathon	UM Hackathon 2026 - Domain 1: AI Agentic Workflow Automation
Team	Dang Wangi 1. Nur Alia Syahira Binti Nor Azam 2. Muhammad Khaidhir Bin Mohd Razin 3. Abdul Hafiz Bin Mohd Noor Azman 4. Nur Farah binti Ahmad Nazri 5. Fakhrul Azim Bin Ahmed Mardzukie
Version	1.0
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AI Model	GLM via Z.AI API and OpenAI
Demo Format	Live web app + slide deck + pre-recorded fallback video
Build Window	10 days full-team

1. Executive Summary

Problem in one sentence: Malaysian construction projects bleed RM 5.75B–9.5B annually due to manual schedule tracking, uncoordinated permit submissions, and reactive quality management, all solvable with a stateful AI agent pipeline that reads messy documents, remembers project state, and acts without a human in the loop.

Buildora is a five-agent AI orchestration system that automates the two biggest manual bottlenecks in Malaysian construction: project workflow coordination and government permit compliance. Built on GLM (Z.AI), it ingests unstructured construction documents, tracks project health statefully across agents, auto-completes CIDB permit applications, and generates actionable reports, without a human in the loop for routine tasks.

The combined GDP uplift potential of digitalising this sector is estimated at RM 4.1B–8.0B/yr.

2. Agent Definitions

Buildora operates five specialised agents, each with a clearly scoped responsibility. The agents are chained through a LangGraph orchestrator, sharing state via Firebase Realtime Database.

	Agent	Role	Key Input	Key Output
A	Document Reader	Receives uploaded documents (PDF, DOCX, images) from the frontend. Parses them using pdfplumber, PyMuPDF, or Tesseract OCR (3-stage fallback). Calls GLM to extract 15+ structured fields. Stores documents in Firebase Storage and writes extracted state to Firebase Realtime Database.	PDF / DOCX / image uploaded by PM	15+ structured fields written to Firebase
B	Schedule Monitor	Reads project state written by Agent A. Monitors milestone submissions from the PM via the milestone form. Detects schedule delays (>3 days beyond baseline) and cost overruns (>8% variance). Computes project health score. Triggers Agent D to send alerts on escalation.	Project state from Firebase + PM milestone form submissions	Escalation flags in Firebase; triggers Agent D
C	CIDB Compliance	Reads project state from Firebase. Validates the project against the CIDB BISQ checklist (cidb_bisq.json). Returns a compliance score (%), list of passed/failed items, and pre-filled form fields.	Project state from Firebase + CIDB BISQ knowledge base	Compliance score, gap list, pre-filled form JSON
D	Alert & Notification	Triggered by Agent B on escalation events. Sends formatted Telegram notifications to the configured project coordinator. Drafts alert messages listing delay details, cost variance, or compliance gaps. Handles scheduled reminders.	Escalation flags and alert data from Agent B	Telegram messages sent within 10 seconds
E	Report Generator	Aggregates outputs from Agents A, B, and C (read from Firebase). Generates a branded PDF project report using ReportLab and an XLSX cost tracker using openpyxl. Supports BM and English output via GLM translation.	All agent outputs from Firebase	Branded PDF report + XLSX cost tracker

3. Problem Statement

3.1 Industry Pain Points

Data sourced from DOSM, CIDB Construction 4.0 benchmarks, and contractor field research.

Pain Point	Current State	Quantified Cost	Agent	Priority
Schedule delays	PMs track milestones in WhatsApp/Excel	74% of projects delayed; average 42% beyond schedule	B	Critical
Cost overruns	No real-time variance flagging	5–10% of contract value lost per project	B	Critical
Permit resubmissions	Forms filled manually; wrong fields, missing docs	60–70% resubmission rate; 2–6 week average delay	C	High
Rework from defects	Inspection sheets on paper or WhatsApp photos	15–25% of project cost in rework	A + B	High
PM admin overhead	Emails, spreadsheets, manual report compilation	~40% of PM time non-productive	E	High
Subcontractor disputes	Payment tracking in Excel; verbal agreements	30–40 day payment delays; project stalls	B	Medium

3.2 Why Current Tools Fail

- Generic PM tools (Trello, Asana) have no construction-domain knowledge and no document parsing
- International tools (Procore, Autodesk) do not support Bahasa Malaysia forms or CIDB G1–G7 grading
- CIDB e-PERMIT system requires manual form completion with no pre-submission validation
- No tool connects schedule tracking, permit status, and quality inspection in one stateful pipeline
- Malaysian-specific constraints (BM forms, CIDB codes, local authority variance) not handled by international software

4. Solution Overview

4.1 Core Architecture Principle

The real product is a single source of truth for construction project state, with AI filling it in automatically. The five agents are the mechanism, the shared Firebase Realtime Database is the product.

LangGraph Orchestrator → Agent A → Agent B → Agent C (conditional) → Agent D → Agent E → Firebase + Telegram + PDF/XLSX

4.2 Data Flow — Full Pipeline

Step	Action
1	PM uploads contract PDF + site plan via React drag-and-drop UI
2	FastAPI receives file → stores in Firebase Storage → triggers LangGraph orchestrator
3	Agent A: pdfplumber/PyMuPDF/Tesseract reads document; GLM extracts 15+ fields
4	Agent A writes structured JSON to Firebase Realtime Database (project state created)
5	Agent B reads Firebase state → checks milestone baseline → flags delays >3 days or cost variance >8%
6	Agent B triggers Agent D on escalation → Telegram alert sent within 10s
7	PM clicks Prepare Permit Package → Agent C reads project state + CIDB BISQ knowledge base (conditional)
8	Agent C returns compliance score, gap list, pre-filled form fields
9	PM clicks Generate Report → Agent E aggregates all Firebase state → branded PDF + XLSX in <10s
10	Frontend displays compliance score + downloadable PDF + XLSX + Telegram notification log

5. Functional Requirements

5.1 Must-Have

Feature	Acceptance Criterion	Agent	Priority
Document upload (PDF/DOCX/image)	Agent A extracts ≥10 fields with ≥80% accuracy on test docs	A	P0
Project state persistence	State survives server restart (Firebase persistence enabled)	A	P0
Milestone tracking form	PM submits update; system flags delay vs baseline	B	P0
Delay & cost alert	Agent B detects >3 day delay or >8% cost variance; triggers Agent D	B + D	P0
Telegram notification	Agent D sends formatted alert within 10 seconds of trigger	D	P0
PDF report generation	Agent E produces valid, readable PDF in <10 seconds	E	P0
Live agent trace panel in UI	Panel shows which agent is active — P0 demo hook	All	P0
Permit compliance check	Agent C returns CIDB checklist with pass/fail — conditional Day 9	C	P0*
XLSX cost tracker output	Valid Excel file with correct sheet structure	E	P1
BM / English toggle on reports	GLM handles BM output for permit form fields	E	P1

5.2 Should-Have

- Multi-project dashboard (list view + per-project detail)
- Drag-and-drop document upload with progress indicator
- Chatbot Assistant (conversational support interface allows users to submit natural language queries about their active project or about CIDB regulatory requirements)
- Demo mode with pre-loaded Malaysian construction project data

5.3 Out of Scope

- Mobile app (web responsive is sufficient for judges)
- Real CIDB API integration, mock the CIDB submission endpoint; acknowledge in Q&A
- Payment processing for subcontractors
- Live IoT sensor data ingestion

6. Technical Architecture

6.1 Tech Stack

Layer	Technology	Rationale
LLM / AI	GLM via Z.AI API	Required by competition; BM + English support
Agent Orchestration	LangGraph (Python)	Native stateful multi-agent graphs; Firebase checkpointer support
Backend API	FastAPI (Python 3.11+)	Async, fast, auto-docs; integrates cleanly with LangGraph
Frontend	React + Vite + Tailwind CSS	Team has web experience; fast to build; polished for demo
PDF Parsing	pdfplumber + PyMuPDF	pdfplumber for text-heavy docs; PyMuPDF for mixed/scanned
OCR Fallback	Tesseract via pytesseract	Scanned drawings and WhatsApp photo docs; BM language pack
Document Output	ReportLab (PDF) + openpyxl	Programmatic; no LibreOffice dependency; demo-stable
State Store	Firebase Realtime Database	Shared state layer for all inter-agent communication
Database	Firebase Firestore	Project records, audit logs, user data
File Storage	Firebase Storage	Uploaded PDFs/images; presigned URL delivery to frontend
Alerts	Telegram Bot API (primary)	Zero approval delay
Containerisation	Docker + Docker Compose	One-command demo setup; no environment mismatch on demo day
Hosting (demo)	Railway / Render + keep-warm cron	Public URL for judges; auto-deploy from GitHub
Testing	pytest + pytest-asyncio	Unit tests for all core agent functions — mandatory from Day 1

6.2 AI Model & Prompt Design

Aspect	Detail
Model	GLM-4-Flash via Z.AI API and OpenAI
Prompting Strategy	Structured few-shot prompting for Agent A field extraction. Multi-step agentic prompting for Agent C compliance reasoning. Zero-shot for Agent E report summarisation.
Context Handling	Documents exceeding 4,000 tokens are chunked by page. Each chunk is processed independently; results are merged. Tables extracted separately via pdfplumber.
Fallback Behavior	3 retries on GLM timeout or malformed JSON response. On third failure: flag document for manual review, continue pipeline with partial state. Pre-cached responses used in demo mode.
BM Support	Tesseract BM language pack for scanned Bahasa Malaysia forms. GLM handles BM output for permit form fields and report generation.

7. User Stories & Use Cases

7.1 Target Users

Stakeholder	Role	Primary Need
Project Manager (PM)	Submits documents, reviews dashboard, receives alerts	Zero-effort project state tracking and early warning on delays
Site Supervisor	Updates milestone progress via mobile web	Quick milestone submission without paperwork
Compliance Officer	Reviews CIDB compliance score before submission	Accurate gap list before visiting PBT counter
QA / Audit Team	Reviews generated reports and agent traces	Auditable AI reasoning trail and downloadable PDF report

7.2 Core User Stories

US-01	As a PM, I want to upload a contract PDF so that the system extracts all key project fields automatically without manual entry.
US-02	As a PM, I want to receive a Telegram alert when a milestone is overdue by more than 3 days so that I can take corrective action immediately.
US-03	As a PM, I want to see a CIDB compliance score before submitting a permit application so that I avoid a costly resubmission.
US-04	As a PM, I want a branded PDF report generated in under 10 seconds so that I can share project status with stakeholders without manual compilation.
US-05	As a compliance officer, I want to see exactly which CIDB checklist items are missing so that I can prepare the correct documents before visiting the PBT counter.
US-06	As a site supervisor, I want to submit milestone updates via a simple form so that the project timeline stays current without email chains.

8. Risk Register & Mitigations

Risk	Severity	Mitigation
GLM API latency >6s per call	HIGH	Pre-cache GLM responses for demo on Day 12. Implement async streaming. Demo must use cached responses only.
GLM accuracy <70% on BM PDFs	HIGH	Day 0 spike: test GLM against 3 real Malaysian construction PDFs before architecture is locked. If <70%: adjust prompts and retest.
Firebase state handoff bugs	HIGH	Mandatory Day 2 gate: Agent A → Agent B handoff demonstrated working before Phase 2. Fallback: sequential pipeline with explicit state dict.
CIDB knowledge base incomplete	HIGH	Manually digitise CIDB BISQ fields before Day 7. Construction professional review required. Day 9 go/no-go: cut Agent C if unconfirmed.
Zero unit tests	HIGH	pytest configured by Day 1. Coverage: Agent A ≥80% accuracy, Agent B delay logic, Agent C scoring. No PR merges without green tests.
Demo cold start >30s	MED	Keep-warm cron pings server every 4 minutes. Alternative: local Docker + ngrok public URL.
Firebase data loss between restarts	LOW	Named Docker volumes only. Demo-data seed script reloads test project in <60 seconds.
ReportLab PDF output unprofessional	MED	0.5 days for PDF template design..

9. Non-Functional Requirements

Category	Requirement	Target
Performance	GLM extraction latency per call	<6 seconds (pre-cache if exceeded)
Performance	PDF report generation time	<10 seconds end-to-end
Performance	Full pipeline (upload → PDF)	<15 seconds on demo scenario
Performance	Telegram alert delivery	<10 seconds from trigger
Reliability	Firebase state persistence	Named Docker volumes; survives server restart
Reliability	GLM failure handling	3 retries → graceful degradation → manual review flag
Reliability	Demo fallback	Pre-cached GLM responses + pre-recorded fallback video
Usability	Setup time for judges	git clone → docker compose up → working demo
Usability	Document onboarding	PM onboards a project in <10 minutes
Security	API keys	Never committed to repo; loaded from .env only
Localisation	Language support	BM and English — GLM handles both
Testability	Test runner	pytest + pytest-asyncio configured in pyproject.toml
Testability	CI gate	GitHub Actions runs pytest on every push to main

10. Assumptions & Constraints

10.1 Assumptions

- Users have stable internet connections on their devices during document upload and milestone submission
- Construction documents are in PDF, DOCX, or common image formats (JPEG, PNG)
- Target users (G7 contractors) already submit CIDB documentation — highest-frequency permit submitters
- GLM-4-Flash is capable of structured field extraction from Malaysian construction documents at $\geq 70\%$ accuracy
- Firebase Realtime Database is sufficient as the inter-agent shared state store within the demo window

10.2 Technical Constraints

- Token cost per session: estimated $\approx 4,000$ – $8,000$ tokens for full pipeline. Mitigation: fixed-length extraction prompts; chunking large documents; caching repeated queries for 24 hours in demo mode
- Maximum document input size: 20 pages / $\approx 8,000$ tokens before chunking is applied
- Agent C compliance check is advisory only — it does not legally certify permit submissions
- Real CIDB e-PERMIT API integration is out of scope; the CIDB submission endpoint is mocked

10.3 LLM Cost Constraint

Item	Detail
Estimated tokens per session	$\sim 4,000$ – $8,000$ (extraction + compliance + report summary)
Cost management decision	Fixed-length extraction prompts; chunking; 24-hour response caching in demo mode
Demo mode	All GLM responses pre-cached by Day 8 — zero API cost on demo day

11. Implementation Timeline (10 Days)

Phase	Days	Theme	Key Deliverables
0	0–1	Foundation + GLM Spike	GLM validation (3 real BM PDFs). Firebase A→B state handoff. Docker stack live. pytest configured. Figma wireframes approved.
1	2–3	Agent A	Upload endpoint + Agent A extraction (≥10 fields, ≥80%). Firebase write. Drag-and-drop UI. test_extract_fields.py green.
2	4–5	Agent B + D	Milestone form. Delay + cost detection. Agent D Telegram alert (<10s). Health score on dashboard. Delay/cost tests green.
3	6–7	Agent C (Conditional)	CIDB JSON complete + professional review. Agent C compliance check. Day 9 go/no-go gate. test_compliance_check.py green.
4	8	Agent E + Integration	Agent E PDF template + XLSX. All agents wired E2E. Agent trace panel in UI. test_pipeline_e2e.py green.
5	9	Demo Hardening	GLM responses pre-cached. Fallback video recorded. Demo mode with realistic data. 5+ rehearsals. Projector test. All 9 tests were green.
6	10	Submission	Final deployment live. Public URL warm. Slide deck uploaded. Repo clean. Submission package complete.

12. Glossary

Term	Definition
CIDB	Construction Industry Development Board Malaysia — regulates contractor registration and building permits
BISQ	CIDB Building Industry Standard Quality system — used for permit compliance checklists
PPK	Perakuan Pendaftaran Kontraktor — CIDB Local Contractor Registration Certificate
PBT	Pihak Berkuasa Tempatan — local authority (e.g. DBKL, MBPJ) that issues development orders
BM	Bahasa Malaysia — national language; required on official permit submissions
GLM	General Language Model — the Z.AI LLM required by the UM Hackathon 2026
LangGraph	Python library for building stateful multi-agent AI graphs
Firebase	Google cloud platform used for Realtime Database (shared state), Firestore (records), and Storage (files)
DOSM	Department of Statistics Malaysia — source for construction GDP and sector figures
G7	CIDB contractor grade — government-registered contractors eligible for unlimited contract value
SHASSIC	Safety and Health Assessment System in Construction — CIDB safety rating tool
LoA	Letter of Award — binding document that triggers insurance submission (within 14 days)
FastAPI	Python async web framework — serves as the API layer between React frontend and LangGraph agents